

Weekly Status Report

E-Pharma Management System — Phase 4

Employee	Jainish Nikul Shah	Designation	Software Engineer
Project	Techmentee — E-Pharma Management System	Coordinator	Vaishali Pujari
Reporting Period	Phase 4 — 03 August 2026 – 07 August 2026	Report Date	7 August 2026
Phase	Phase 4 — Admin ERP & Reporting	Overall Status	Completed (ahead of schedule)

1. Summary

All Phase 4 objectives — administrative reporting, catalog and content management, and an SMS/email notification gateway — were completed and verified end-to-end. A key design decision this phase was to implement the SMS/email gateway as an additional consumer on the Kafka notification stream introduced in Phase 3, rather than as inline code: every notification is now both persisted and delivered, and new delivery channels can be added without touching business logic. Reporting was built with plain aggregation queries and CSS charts to avoid an unnecessary charting dependency, and the managed catalog was kept advisory so existing free-text data continues to work.

2. Tasks Completed This Phase

Task (Phase 4 scope)	Status	Remarks
Revenue / consultation / order reports	Completed	Admin reporting endpoint with aggregations (revenue by day, top medicines, revenue by pharmacy, order & appointment status, consultation totals), rendered as CSS bar charts — no charting dependency added
SMS / email notification gateway	Completed	Implemented as a second consumer on the existing Kafka notifications topic (fan-out: each notification is persisted AND delivered). Mock delivery by default; NOTIFY_CHANNELS + Twilio/SES to go live
Category & specialty management	Completed	Admin-managed taxonomy; surfaced as datalist suggestions in medicine and registration forms. Advisory by design — free text still accepted, so existing data and flows are unaffected
CMS pages (FAQs, Terms, Privacy)	Completed	Admin-editable pages stored in the database; linked from the landing footer with a read-only viewer for patients
Testing & documentation	Completed	End-to-end test suite grown to 45 assertions (reports, taxonomy management, CMS editing, RBAC); README, analysis document and development plan updated

3. Deliverables

- Admin reporting endpoint and dashboard tab — revenue, top medicines, per-pharmacy revenue, consultation and status breakdowns
- SMS/email delivery service subscribing to the Kafka notifications topic (fan-out delivery)
- Catalog management (categories & specialties) with datalist-backed form suggestions
- Admin-editable CMS pages (FAQ / Terms / Privacy) with public landing-footer links
- Automated end-to-end test suite expanded to 45 passing assertions; documentation updated

4. Key System-Design Decisions

- SMS/email delivery as a Kafka consumer (fan-out), not inline — producers stay decoupled from delivery, and channels scale independently
- Reporting via SQL aggregation + CSS charts — no charting library added for a handful of bar charts
- Managed catalog is advisory (datalist), not a hard foreign-key constraint — avoids breaking existing free-text categories/specialties and any in-flight data
- CMS stored in the database (admin-editable) rather than static files, so content changes need no redeploy

5. Plan for Next Phase (Phase 5 — Performance & Security)

- Migrate persistence from SQLite to PostgreSQL for production scale
- Encryption at rest for prescriptions/medical records; audit logging; data-retention policy
- Load testing and an accessibility pass ahead of deployment

6. Issues / Blockers

None blocking. For production readiness, confirmation is requested on the target hosting environment (for the PostgreSQL migration and a managed Kafka cluster) and on the SMS/email provider to license (Twilio / AWS SES) for live notification delivery.

Prepared by: Jainish Nikul Shah, Software Engineer

Submitted to: Vaishali Pujari, Project Coordinator